

Mode 2 — Square-BFP (centered / opt)

DP16 (8×8) → DP16 (8×4) → DP8 (8×4) → block exponents → DP8 (4×4) → centered aligned
add-and-square

Step 1 — DP16 (8×8)

$$\text{DP16 (8×8)} = \sum_{i=0}^{15} a_i b_i$$

Step 2 — DP16 (8×4)

$$\text{DP16 (8×8)} = \underbrace{2^4 \sum_{i=0}^{15} a_i b_i^{hi}}_{\text{DP16 (8×4)}} + \underbrace{\sum_{i=0}^{15} a_i b_i^{lo}}_{\text{DP16 (8×4)}}$$

Step 3 — DP8 (8×4)

$$\text{DP16 (8×8)} = \underbrace{2^4 \sum_{i=0}^7 a_i b_i^{hi}}_{\text{DP8 (8×4)}} + \underbrace{2^4 \sum_{i=8}^{15} a_i b_i^{hi}}_{\text{DP8 (8×4)}} + \underbrace{\sum_{i=0}^7 a_i b_i^{lo}}_{\text{DP8 (8×4)}} + \underbrace{\sum_{i=8}^{15} a_i b_i^{lo}}_{\text{DP8 (8×4)}}$$

Step 4 — Block exponents (one exponent block per DP8 (8×4))

$$\left(2^4 \sum_{i=0}^7 a_i b_i^{hi} + \sum_{i=0}^7 a_i b_i^{lo}, E_0 \right), \left(2^4 \sum_{i=8}^{15} a_i b_i^{hi} + \sum_{i=8}^{15} a_i b_i^{lo}, E_1 \right) \longrightarrow (M, E)$$

$$M = 2^{-\delta_0} \left(2^4 \sum_{i=0}^7 a_i b_i^{hi} + \sum_{i=0}^7 a_i b_i^{lo} \right) + 2^{-\delta_1} \left(2^4 \sum_{i=8}^{15} a_i b_i^{hi} + \sum_{i=8}^{15} a_i b_i^{lo} \right)$$

$E_k = E_{a,k} + E_{b,k}$; $E = \max(E_0, E_1)$, $\delta_k = E - E_k$; truncating shifts, on block sums only; $\text{DP16 (8×8)} = M \cdot 2^E$.

Step 5 — DP8 (4×4)

$$\begin{aligned} M = 2^{-\delta_0} & \left(\underbrace{2^8 \sum_{i=0}^7 a_{H,i} b_i^{hi}}_{\text{DP8 (4×4)}} + \underbrace{2^4 \sum_{i=0}^7 a_{L,i} b_i^{hi}}_{\text{DP8 (4×4)}} + \underbrace{2^4 \sum_{i=0}^7 a_{H,i} b_i^{lo}}_{\text{DP8 (4×4)}} + \underbrace{\sum_{i=0}^7 a_{L,i} b_i^{lo}}_{\text{DP8 (4×4)}} \right) \\ & + 2^{-\delta_1} \left(\underbrace{2^8 \sum_{i=8}^{15} a_{H,i} b_i^{hi}}_{\text{DP8 (4×4)}} + \underbrace{2^4 \sum_{i=8}^{15} a_{L,i} b_i^{hi}}_{\text{DP8 (4×4)}} + \underbrace{2^4 \sum_{i=8}^{15} a_{H,i} b_i^{lo}}_{\text{DP8 (4×4)}} + \underbrace{\sum_{i=8}^{15} a_{L,i} b_i^{lo}}_{\text{DP8 (4×4)}} \right) \end{aligned}$$

Step 6 — Centered aligned add-and-square (subtract 8 from unsigned nibbles a_L, b^{lo} ; per DP8 (4×4) add $c = 0/512/2048$ for 0/1/2 unsigned nibbles)

$$\begin{aligned} M = 2^{-\delta_0} & 2^8 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i} + b_i^{hi})^2 - \sum_{i=0}^7 a_{H,i}^2 - \sum_{i=0}^7 (b_i^{hi})^2 \right)}_{\text{DP8 (4×4)}} \\ & + 2^{-\delta_0} 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 ((a_{L,i} - 8) + b_i^{hi})^2 - \sum_{i=0}^7 (a_{L,i} - 8)^2 - \sum_{i=0}^7 (b_i^{hi} - 8)^2 + 512 \right)}_{\text{DP8 (4×4)}} \\ & + 2^{-\delta_0} 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i} + (b_i^{lo} - 8))^2 - \sum_{i=0}^7 (a_{H,i} - 8)^2 - \sum_{i=0}^7 (b_i^{lo} - 8)^2 + 512 \right)}_{\text{DP8 (4×4)}} \\ & + 2^{-\delta_0} \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 ((a_{L,i} - 8) + (b_i^{lo} - 8))^2 - \sum_{i=0}^7 (a_{L,i} - 16)^2 - \sum_{i=0}^7 (b_i^{lo} - 16)^2 + 2048 \right)}_{\text{DP8 (4×4)}} \end{aligned}$$

$$\begin{aligned}
& + 2^{-\delta_1} 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{H,i} + b_i^{hi})^2 - \sum_{i=8}^{15} a_{H,i}^2 - \sum_{i=8}^{15} (b_i^{hi})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{-\delta_1} 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{L,i}-8) + b_i^{hi})^2 - \sum_{i=8}^{15} (a_{L,i}-8)^2 - \sum_{i=8}^{15} (b_i^{hi}-8)^2 + 512 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{-\delta_1} 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{H,i} + (b_i^{lo}-8))^2 - \sum_{i=8}^{15} (a_{H,i}-8)^2 - \sum_{i=8}^{15} (b_i^{lo}-8)^2 + 512 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{-\delta_1} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{L,i}-8) + (b_i^{lo}-8))^2 - \sum_{i=8}^{15} (a_{L,i}-16)^2 - \sum_{i=8}^{15} (b_i^{lo}-16)^2 + 2048 \right)}_{\text{DP8 (4}\times\text{4)}}
\end{aligned}$$

Step 7 — All components

$$\begin{aligned}
\text{PE} &= 2^{-\delta_0} \left(2^8 \sum_{i=0}^7 (a_{H,i} + b_i^{hi})^2 + 2^4 \sum_{i=0}^7 ((a_{L,i}-8) + b_i^{hi})^2 + 2^4 \sum_{i=0}^7 (a_{H,i} + (b_i^{lo}-8))^2 + \sum_{i=0}^7 ((a_{L,i}-8) + (b_i^{lo}-8))^2 \right) \\
&+ 2^{-\delta_1} \left(2^8 \sum_{i=8}^{15} (a_{H,i} + b_i^{hi})^2 + 2^4 \sum_{i=8}^{15} ((a_{L,i}-8) + b_i^{hi})^2 + 2^4 \sum_{i=8}^{15} (a_{H,i} + (b_i^{lo}-8))^2 + \sum_{i=8}^{15} ((a_{L,i}-8) + (b_i^{lo}-8))^2 \right) \\
\alpha &= 2^{-\delta_0} \left(2^8 \sum_{i=0}^7 a_{H,i}^2 + 2^4 \sum_{i=0}^7 (a_{L,i}-8)^2 + 2^4 \sum_{i=0}^7 (a_{H,i}-8)^2 + \sum_{i=0}^7 (a_{L,i}-16)^2 \right) \\
&+ 2^{-\delta_1} \left(2^8 \sum_{i=8}^{15} a_{H,i}^2 + 2^4 \sum_{i=8}^{15} (a_{L,i}-8)^2 + 2^4 \sum_{i=8}^{15} (a_{H,i}-8)^2 + \sum_{i=8}^{15} (a_{L,i}-16)^2 \right) \\
\beta &= 2^{-\delta_0} \left(2^8 \sum_{i=0}^7 (b_i^{hi})^2 + 2^4 \sum_{i=0}^7 (b_i^{hi}-8)^2 + 2^4 \sum_{i=0}^7 (b_i^{lo}-8)^2 + \sum_{i=0}^7 (b_i^{lo}-16)^2 \right) \\
&+ 2^{-\delta_1} \left(2^8 \sum_{i=8}^{15} (b_i^{hi})^2 + 2^4 \sum_{i=8}^{15} (b_i^{hi}-8)^2 + 2^4 \sum_{i=8}^{15} (b_i^{lo}-8)^2 + \sum_{i=8}^{15} (b_i^{lo}-16)^2 \right) \\
C &= 2^{-\delta_0} (0 + 8192 + 8192 + 2048) + 2^{-\delta_1} (0 + 8192 + 8192 + 2048) = 2^{-\delta_0} \cdot 18432 + 2^{-\delta_1} \cdot 18432 \\
M &= \frac{1}{2} (\text{PE} - \alpha - \beta + C), \quad E = \max(E_0, E_1)
\end{aligned}$$